MC 5.2 (a), 6.1 (b), 6.2 (a)

$$\begin{cases} A = \underbrace{\varepsilon + B1}_{T} + A\underbrace{0}_{S} \\ B = A1 \\ C = B0 + C1 \\ D = C0 + D1 \end{cases} \Rightarrow \begin{cases} A = (\varepsilon + B1)0^* \\ B = (\varepsilon + B1)0^*1 = \underbrace{0^*1}_{T} + B1\underbrace{0^*1}_{S} \\ - \\ - \end{cases}$$

$$\Rightarrow \begin{cases} - \\ B = 0^*1(10^*1)^* \\ C = \underbrace{0^*1(10^*1)^*0}_{T} + C\underbrace{1}_{S} \\ - \end{cases} \Rightarrow \begin{cases} - \\ - \\ C = 0^*1(10^*1)^*0^*1^* \\ D = C0 + D1 \Rightarrow D = C0^*1^* \end{cases}$$

$$D = 0^*1(10^*1)0^*1^*0^*1^*$$

R: A linguagem aceite pelo autómato é dada pela expressão regular

$$B + D$$

5.2(c)

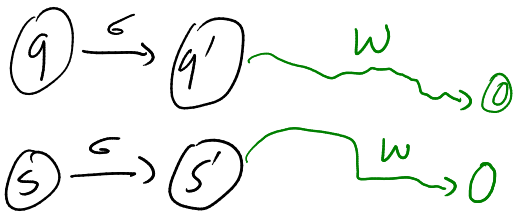
$$\begin{cases} A = \varepsilon + A1 + C0 \\ B = A0 + B1 \\ C = B0 + C1 \end{cases} \Rightarrow \begin{cases} A = \varepsilon + A1 + A001^*0 + B101^*0 \\ — \\ C = B01^* = A001^* + B101^* \end{cases}$$

$$\Leftrightarrow \begin{cases} A = \varepsilon + B101^*0 + A(1 + 001^*0) \\ \end{cases} \Rightarrow \begin{cases} A = (\varepsilon + B101^*0)\cdot(1+001^*0)^* \\ B = (\varepsilon + B101^*0)(1+001^*0)^*0 + B1 \\ — \end{cases}$$

$$\Leftrightarrow \begin{cases} B = (1+001^*0)^*0 + B(101^*0 + 1) \end{cases} \Rightarrow \begin{cases} B = (1+001^*0)^*0(101^*0+1)^* \end{cases}$$

R: A linguagem decidida pelo autómato é $\mathcal{L}(B)$

6.1(b)



$Dist(q,s) \overset{def}{\iff} \exists w \in \Sigma^* \; \exists q', s' \in Q$

$q \xrightarrow{w} q'$

$\wedge \; s \xrightarrow{w} s'$

$\wedge \; |\{q', s'\} \cap F| = 1$

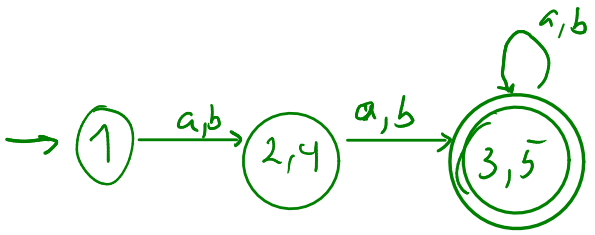
$q \in F, \; s \notin F \Rightarrow Dist(q,s)$

$q \xmapsto{\sigma} q' \quad \delta(q,\sigma) = q'$

$s \xmapsto{\sigma} s' \quad \delta(s,\sigma) = s' \Rightarrow Dist(q,s)$

$Dist(q',s')$

1					
2	X_1				
3	X_0	X_0			
4	X_1		X_1		
5	X_0	X_0		X_0	
	1	2	3	4	5



Iteração 1

$(1,2) \xrightarrow{a} (2,3)$

$(1,4) \xrightarrow{b} (4,5)$

$(2,4)$ x

$(3,5)$ x

Iteração 2

$(2,4)$ x

$(3,5)$ x

$R = 01^* + \underbrace{(0+11)^*\cdot 0}_{K}$

$D_0 K = D_0((0+11)^*)\cdot 0 + D_0(0)$

$= D_0(0+11)(0+11)^*\cdot 0 + \varepsilon$

$= (0+11)^*0 + \varepsilon = K + \varepsilon =: U$

$D_1 K = 1(0+11)^*0 = 1K =: T$

$D_0 R = 1^* + K + \varepsilon =: S$

$D_1 R = D_1 K = 1K =: T$

$D_0 S = K + \varepsilon =: U$

$D_1 S = 1^* + 1K =: V$

$D_0 T = \emptyset$

$D_1 T = K$

$D_0 U = D_0 K = U$

$D_1 U = D_1 K = T$

$D_0 V = \emptyset$

$D_1 V = 1^* + K =: W$

$D_0 W = D_0 K = U$

$D_1 W = 1^* + D_1 K = V$

		0	1
→	R	S	T
*	S	U	V
	T	$\emptyset$	K
*	U	U	T
*	V	$\emptyset$	W
	K	U	T
*	W	U	V
	$\emptyset$	$\emptyset$	$\emptyset$

R							
S	X_0						
T		X_0					
U	X_0		X_0				
V	X_0		X_0				
K		X_0		X_0	X_0		
W	X_0		X_0			X_0	
$\emptyset$		X_0		X_0	X_0		X_0
	R	S	T	U	V	K	W	$\emptyset$

$D_0 V = \emptyset$

$D_1 V = 1^* + K =: W$

$D_0 W = D_0 K = U$

$D_1 W = 1^* + D_1 K = V$

		0	1
→	R	SW	T
*	SW	U	V
	T	$\emptyset$	K
*	U	U	T
*	V	$\emptyset$	SW
	K	U	T
	$\emptyset$	$\emptyset$	$\emptyset$

Iteração 1

$(R,T) \xrightarrow{0} (S,\emptyset)$

(R,K) x

$(R,\emptyset) \xrightarrow{0} (S,\emptyset)$

$(S,U) \xrightarrow{1} (V,T)$

$(S,V) \rightarrow (U,\emptyset)$

(S,W) x

$(T,K) \rightarrow (\emptyset,U)$

$(T,\emptyset)$ x

$(U,V) \xrightarrow{0} (U,\emptyset)$

$(U,W) \xrightarrow{1} (T,V)$

$(V,W) \xrightarrow{0} (\emptyset,U)$

$(K,\emptyset) \xrightarrow{0} (U,\emptyset)$

Iteração 2

$(R,K) \xrightarrow{0} (S,U)$

(S,W) x

$(T,\emptyset) \rightarrow (K,\emptyset)$

Iteração 3

(S,W) x

Não foram encontrados novos pares de estados distinguíveis

	R	S	T	U	V	K	W	$\emptyset$
S	X_0							
T	X_1	X_0						
U	X_0	X_1	X_0					
V	X_0	X_1	X_0	X_1				
K	X_2	X_0	X_1	X_0	X_0			
W	X_0		X_0	X_1	X_1	X_0		
$\emptyset$	X_1	X_0	X_2	X_0	X_0	X_1	X_0	

$R = (a+b)^* \underbrace{(a^* ba + a^* bba)^*}_{K}$

$D_a K = (D_a(a^* ba) + D_a(a^* bba))\, K$

$= (a^* ba + a^* bba)\, K =: L$

$D_b K = (a + ba)\, K =: M$

$D_a R = (a+b)^* K + L =: S$

$D_b R = (a+b)^* K + M =: T$

$D_a S = (a+b)^* K + L = S$

$D_b S = (a+b)^* K + M = T$

$D_a T = (a+b)^* K + L + K =: U$

$D_b T = (a+b)^* K + M + aK =: V$

$D_a U = (a+b)^* K + L = S$

$D_b U = (a+b)^* K + M = T$

$D_a L = (a^* ba + a^* bba)\, K = L$

$D_b L = (a + ba)\, K = M$

$D_a M = K$

$D_b M = aK$

$D_a V = (a+b)^* K + L + K = U$

$D_b V = (a+b)^* K + M + aK = V$

		a	b
→ *	R	S	T
*	S	S	T
*	T	U	V
*	U	S	T
*	V	U	V

$D_b K = D_b(a^* ba + a^* bba) \cdot K$

$D_b(a^* ba) = D_b(a^*)\, ba + D_b(ba) = a$